

## Scalability & Future Plan

|               |                           |
|---------------|---------------------------|
| Date          | 2 <sup>nd</sup> July 2024 |
| Team ID       |                           |
| Project Name  | Rising Waters             |
| Maximum Marks | 1 Mark                    |

### Scalability & Future Plan: Current System Limitations

| S.No | Limitation                   | Impact                                  | Priority |
|------|------------------------------|-----------------------------------------|----------|
| 1    | Uses historical dataset only | Cannot predict using live weather       | High     |
| 2    | Local deployment             | Limited accessibility                   | Medium   |
| 3    | Single prediction model      | Limited comparison with advanced models | Medium   |

### Scalability Plan

| S.No | Scalability Aspect | Current State        | Proposed Upgrade                          |
|------|--------------------|----------------------|-------------------------------------------|
| 1    | User Load          | Supports local users | Deploy on IBM Cloud for multiple users    |
| 2    | Data Storage       | Local CSV dataset    | Cloud database integration                |
| 3    | Performance        | Local Flask server   | Production deployment with Gunicorn/Nginx |
| 4    | Security           | Basic validation     | User authentication and HTTPS             |

### Future Roadmap

| Phase   | Planned Feature                   | Target Timeline | Expected Impact       |
|---------|-----------------------------------|-----------------|-----------------------|
| Phase 1 | IBM Cloud Deployment              | 2 Months        | Online accessibility  |
| Phase 2 | Real-Time Weather API Integration | 3 Months        | Live flood prediction |

|         |                             |          |                             |
|---------|-----------------------------|----------|-----------------------------|
| Phase 3 | SMS & Email<br>Alert System | 4 Months | Early warning notifications |
| Phase 4 | Mobile<br>Application       | 6 Months | Improved accessibility      |